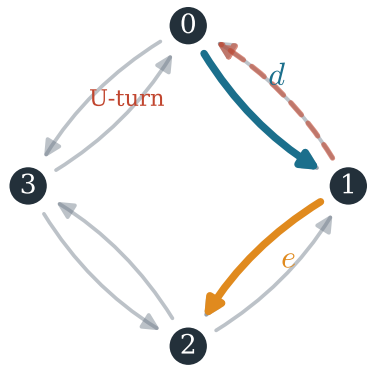


Bass's formula: the $2|E|$ -dimensional non-backtracking determinant, from $|V|$ -dimensional data

$B_{d,e} = 1$: head of d = tail of e , $e \neq d^{\text{symm}}$
($2|E| = 8$ darts)



Bass
→

$$(1 - u^2)^{|V|} \det(I - uB)$$
$$= (1 - u^2)^{|E|} \det(I - uA + u^2(D - I))$$
$$2|E| \text{-dim} \longrightarrow |V| \text{-dim}$$